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Harley-Davidson: chasing a new generation of customers

How to win young riders without losing the Core — a recommendation to the CMO, as seen in 2007–08

Individual report
Applied Problem-Solving: Techniques for Business Decision-Making
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Case: Roese, N. J. & Kompella, M. (2013), Kellogg School of Management, KEL742

EXECUTIVE SUMMARY

Harley should win young riders under its own name, with dark bikes it already makes, and sell Buell

Harley-Davidson enters 2008 as the leader of the US heavyweight market[1] (44.6% share), but its customers are ageing (median age 35 → 47 between 1987 and 2007) and young adults are not replacing them. They rate Harley a "Leadership" brand[2], yet only 16% feel it is for them. The new CMO has about 27 months, a new bike takes four years, and the economy is entering a crisis.

Decision	Recommendation	Main reason
Target	Young adults (18–34), without losing the Core[3]	They decide whether Harley has a next generation; Harley holds only 22% of them
Brand	One Harley-Davidson name (a branded house[4]), not Buell[5]	Young adults already respect the Harley name; Buell carries none of it
Bike	The existing dark models[6] as one line, led by the Iron 883[7] at $7,499	Young adults prefer dark bikes; all are ready within the CMO's window
Price	Lead every message with "from $6,699"	Young adults think a Harley costs $20,000, three times the real entry price
Dealership	Young-adult Garage Parties[8], test rides, younger advisers, Rider's Edge[9] training; piloted first	Intimidating dealerships are the largest barrier
Brand image	Films showing young riders on dark Harleys; a young apparel line with a free test ride	Films like *Easy Rider* built the Boomers' dream; apparel is a cheap first contact
Marketing and data	Outreach[10] from 17% towards 30% of marketing; Core events protected; sales reported by customer segment	Money and data still follow the old customer base
Buell	Freeze investment and sell	The 1125R destroyed $10.5m; keeping Buell is worth −$14.6m; in a crisis every dollar must earn 35%, not 5%

Source: case pp. 1–8 and Exhibits 1–2; author analysis and DCF (Section 5, Appendix H).

Why this works. Every product exists or arrives by month 18, so it fits the CMO's window; it needs under $1 million of new capital, so it holds up in the crisis; and it serves the mission to fulfil dreams "through the experience of motorcycling" for "the general public", while the Core keeps its chrome bikes and rallies.

1 Motorcycles with engines larger than 601cc, mainly cruisers and touring bikes: Harley's main category.
2 Young & Rubicam's BrandAsset Valuator, a brand-strength survey (case Exhibit 1). The "Leadership" quadrant means well known and highly valued.
3 Harley's traditional customers: Caucasian men aged 35 and over, mostly Baby Boomers.
4 One brand covers every product, as at Nike; the alternative, a house of brands, runs separate brands, as at Procter & Gamble.
5 A sport-bike brand made by Harley and sold through its dealers, with no Harley name or design on the bikes.
6 Black, low-chrome, stripped-down Harleys: Nightster, Street Bob, Night Train, Night Rod Special and the soon-to-be-launched CrossBones.
7 A blacked-out Sportster, Harley's lowest-priced family; developed for under $1 million, due January 2009 at $7,499.
8 Small, informal dealer events, so far only for women, to introduce newcomers to motorcycles.
9 Harley's training programme that teaches new riders to ride.
10 Harley's name for the groups it wants to grow: young adults, women, and Hispanic and African-American riders.

PART I · 1 THE PROBLEM

Harley's customers are ageing and the company is not replacing them, which threatens its future sales and its mission

> **Problem statement.** Harley-Davidson's customers are ageing (median age 35 → 47 between 1987 and 2007), and the company is not recruiting enough young riders to replace them. Sales are still strong, with a record year in 2006, but the ageing threatens future sales. The first warning sign is that Harley fell faster than the market in 2007 (−6.1% against −4.1%).

Link to the mission. Harley's mission was: *"We fulfill dreams through the experience of motorcycling, by providing to motorcyclists and to the general public an expanding line of motorcycles and branded products and services in selected market segments."*[11] A company whose purpose is to fulfil dreams for "the general public" with "an expanding line" cannot do so if its riders come from one generation that will age out of riding.

Question	Answer from the case
Who is the decision maker?	Mark-Hans Richer, Harley's first Chief Marketing Officer (July 2007)
What is his strategy?	Win "a new generation of riders and a more diverse customer base, all without losing current customers"
What is the environment?	Market leader, but losing share and entering a recession (Section 2)
What are the constraints?	27-month average CMO tenure against a 4-year product cycle; dealers who see Outreach spending as a cost to the Core
Why does it matter?	Baby Boomers (43–61 in 2007) will age out of riding, and with them most of Harley's buyers
How is success measured?	Young Adult share (22%), "can relate" score of under-35s (16%), median buyer age (47), Core retention

Exhibit 1: Internal status: Harley's strengths sit in its brand; its weaknesses all concern young customers

Strengths	Weaknesses
“Leadership” brand, also among young adults; 44.6% share; 35% gross margin, the highest in the industry; H.O.G. community of 600,000+; dealer network; dark models already on sale	Ageing customer base (median 47); only 16% of under-35s relate to the brand; intimidating dealers; sales data blind to segments; Buell earns 5%
Opportunities	**Threats**
Young adults (Harley holds only 22%); the “black world” taste for dark bikes; sport-bike owners who find their bikes soulless; women and Hispanic riders	Recession and a 20% fall in the share price; cheaper Japanese copies; Boomers ageing out of riding

So what (TOWS): use the brand's strengths to fix its weaknesses with young adults (Part II, Sections 4 and 6), and exit the weakest business, Buell, while the market is threatened (Section 5). Source: case pp. 2–8 and Exhibit 1.

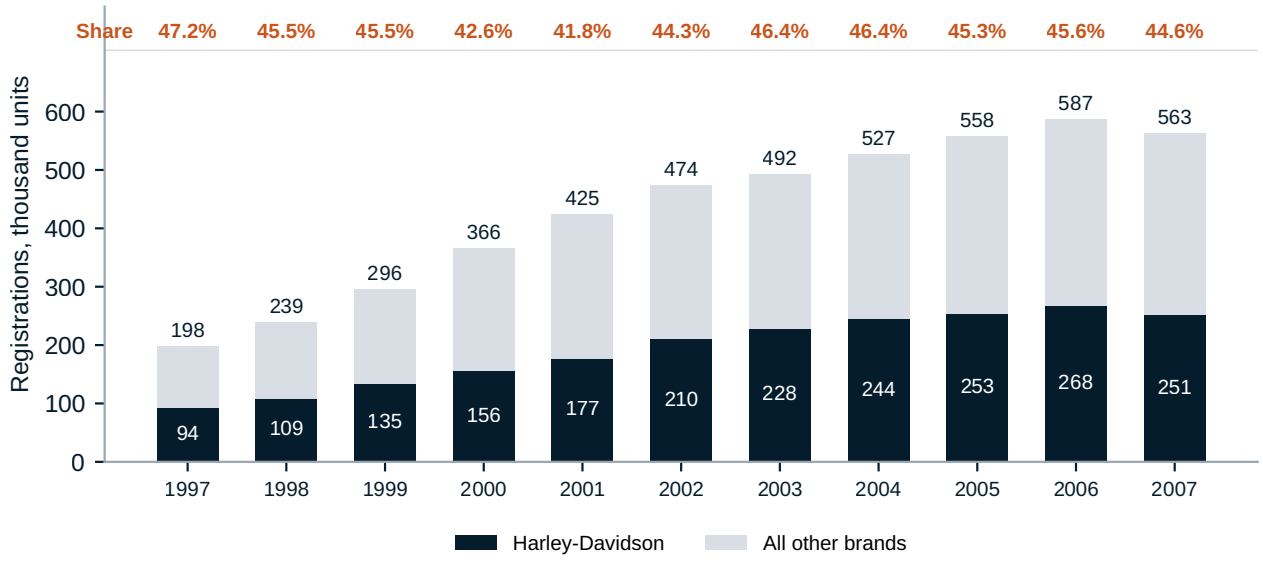
11 Not in the case. Taken from secondary sources reporting Harley-Davidson's mission and values of the period (see Appendix I).

The heavyweight industry is attractive for a brand leader, but it is cooling and Harley has started to lose ground faster than the market

The market almost tripled in 1997–2006, but Harley grew more slowly, and in 2007 it fell faster than the market. The gap is specific to Harley, not just the economy.

Exhibit 2: Harley lost 2.6 points of US heavyweight share over ten years and fell faster than the market in 2007

US registrations of 601cc+ motorcycles, thousand units; Harley share of the total, %



Source: case Exhibit 2 (Motorcycle Industry Council registration data). Harley units calculated as the industry total minus all other brands.

Force	Evidence from the case	Pressure
Rivalry	Suzuki, Honda, Yamaha and Kawasaki hold 47.3% combined and sell copies of Harley's cruiser styles at lower prices and higher speeds	High
Buyer power	Heavyweights are expensive discretionary purchases, and buyers delay them in a downturn	Rising
Substitutes	For young riders, sport bikes (Ducati, Kawasaki); for all, other leisure spending	Medium–high
Threat of entry	Brand heritage, dealer networks and riding communities take decades to build	Low
Supplier power	Not discussed in the case; assumed not decisive for this problem	n/a

Exhibit 3: Rivalry and substitutes are strong, but entry barriers protect the leader (Porter's Five Forces, US heavyweight industry). Source: case pp. 2–3, 8.

Conclusion. The industry is attractive for a brand-led leader, because entry is hard and loyalty high, but it is turning hostile: demand is falling, and for young riders the Japanese compete on price and speed. Harley should compete on its brand and community, not on the rivals' terms.

Young adults know and respect Harley; seven fixable causes stop them from becoming owners

Young Adults (18–34) are the target: the case calls them "the biggest question hanging over the brand", and Harley holds only 22% of them.

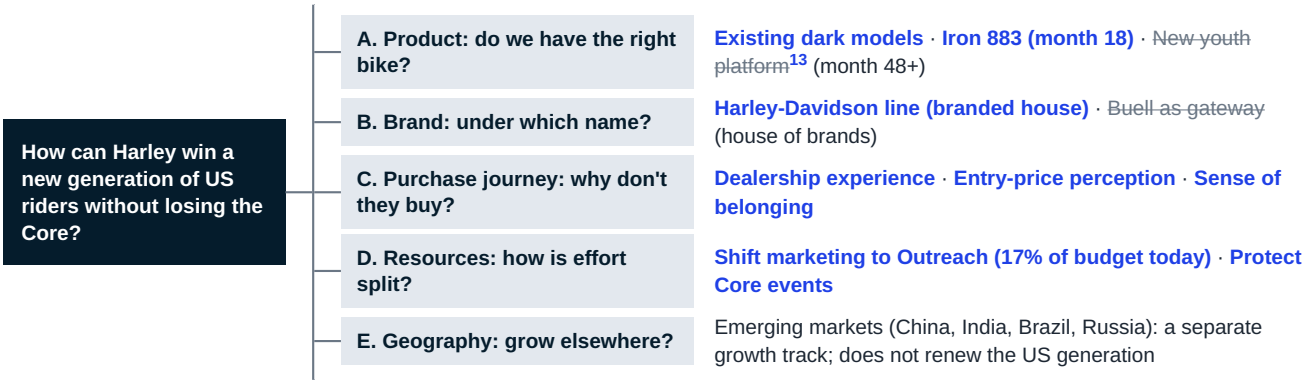
The gap is relevance, not awareness: young adults rate Harley a "Leadership" brand, but do not feel it is for them. Seven causes explain why.

Exhibit 4: Seven causes of the problem, each addressed in Part II

#	Cause	Evidence from the case	Fixed in
Reported by young adults (Harley's 2007 research)			
1	**Intimidating dealerships** (largest barrier)	Young adults find the visit "daunting and intimidating", so interest created by advertising is lost at the point of sale	Section 4
2	**"Haven't seen the right bike"**	They prefer dark, low-chrome, stripped-down bikes, a taste a 2006 European study called the "black world"	Section 4
3	**"May not belong"**	16% of under-35s relate to Harley against 32% of over-35s; riding clubs rarely recruit newcomers	Section 4
4	**Price misperception**	Typical belief: "every Harley costs $20 grand"; actual entry price $6,699	Section 4
Inside Harley			
5	**Marketing focused on the Core**	Only 17% of planned 2007 US marketing spending went to Outreach groups	Section 4
6	**Data that hid young buyers**	Sales were reported by model family[12], not by customer type, so Harley did not notice that young adults were already buying the dark models	Section 4
7	**Reliance on Buell as the way in**	Buell carries no Harley name, earns a 5% gross margin and sells fewer than 5,600 bikes a year in the US	Section 5

Source: case pp. 4–7 and Exhibit 1; grouping by the author.

Exhibit 5: Issue tree: five levers, three prioritised



Blue = prioritised; struck through = rejected. Source: author analysis of the case.

Options rejected. A new youth motorcycle[14] would arrive four years later, 21 months after the average CMO has left; Buell as the gateway does not make the Harley brand younger; emerging markets add growth but do not renew the US customer base. Prioritisation and timeline: Appendix A.

12 Harley groups its bikes into model families with their own frame and engine: Sportster (the lowest-priced), Dyna, Softail, V-Rod and Touring.
13 A completely new motorcycle designed from scratch for young riders, with a new engine and frame. The case says a new product takes at least four years to develop.
14 A completely new model designed from scratch for young riders, with a new engine and frame.

PART II · 4 THE SOLUTION

Win young riders under the Harley name, with dark bikes Harley already makes and a dealership that welcomes them

Branded house (one name) or house of brands (Buell as the youth brand)? Harley's brand equity and the portfolio matrix both favour the Harley name.

Exhibit 6: The Harley name carries what young adults value; Buell does not

Criterion	Branded house: Harley line for young riders	House of brands: Buell as the youth brand
Brand equity with young adults	"Leadership" quadrant in the BAV; associated with transformation, escape and belonging	No Harley branding on the bikes, so none of these associations carry over
Product fit	Dark models young adults already buy proportionally more of	Light sport bikes, competing on speed and handling
Competitive position	Leader; competes on authenticity, where rivals are weak	Fewer than 5,600 US sales against Suzuki's 27,000 in Buell's engine class
Gross margin	About 35%	About 5% (hoped-for 10%)
Path to the core brand	Direct: the rider is already a Harley owner and H.O.G.-eligible	Trade-up from Buell to Harley is an expectation, not proven in the case
Portfolio matrix (BCG)	**Cash cow**: market leader (44.6% share) in a mature market that has stopped growing	**Dog**: under 1% of the US market, a fifth of Suzuki's volume in its class, in a market where it rarely makes a profit

Source: case pp. 5–7 and Exhibit 1.

Exhibit 7: Each cause has an action, a date and a measure

#	Cause	Action	When	Measure (2007 baseline)
1	Intimidating dealerships	Young-adult Garage Parties, test rides, younger advisers, trained staff; piloted in a few dealers first	Pilot months 3–9; scale 9–18	Under-35 visit-to-purchase conversion (not measured today)
2	"Haven't seen the right bike"	One Harley line of the dark models[15], led by the Iron 883 at $7,499	Now; Iron 883 at month 18	Young Adult share (22%); Iron 883 above ~380 bikes, its payback
3	"May not belong"	Rider's Edge with the first bike; H.O.G.[16] rides for new riders; films showing young riders on dark Harleys; a young apparel line with a free test ride	Months 3–18	"Can relate", under-35s: 16% → 24%+
4	Price misperception	Lead every message and showroom with "from $6,699"	Months 3–9	Perceived entry price ($20,000)
5	Marketing focused on the Core	Raise Outreach spending in stages; protect Core rallies and events	Months 9–18	Outreach share (17% → ~30%); Core retention
6	Data that hid young buyers	Report sales by customer segment, not only by model family	Months 0–3	Median buyer age (47) stops rising
7	Reliance on Buell	Freeze investment and run a sale process (Section 5)	Months 0–27	Sale proceeds vs value of keeping

Targets of 24% and ~30% are author assumptions. Full plan, KPIs and the barrier detail: Appendices B–D.

> **Positioning statement.** For young adults who want to escape a constrained routine (*target*), Harley's dark line is the authentic American motorcycle (*what*) that comes with a community to ride with, unlike mass-produced sport bikes (*USP*), so they can become who they want to be, from $6,699 (*benefit*), backed by a century of riding culture and 600,000 H.O.G. members (*evidence*).

15 Nightster, Street Bob, Night Train, Night Rod Special and the soon-to-be-launched CrossBones, a dark Softail model.
16 Harley Owners Group: Harley's own riders' club, over 600,000 members in 2007.

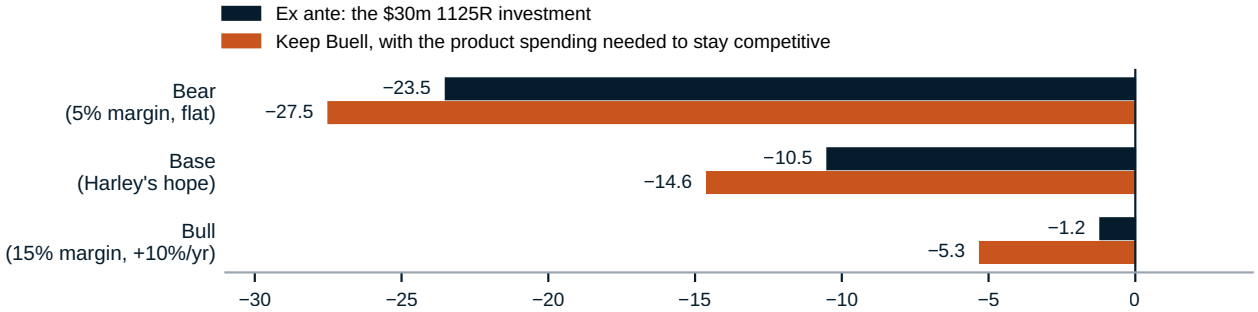
PART II · 5 WHAT TO DO WITH BUELL

The $30 million 1125R destroyed value and keeping Buell loses money: in a crisis, Harley should sell it

Buell carries no Harley name and sells fewer than 5,600 bikes a year in the US. A discounted cash flow (DCF) was built from the case data: Buell's 5% gross margin, the 10% it hopes for, and the $30 million 1125R[17]. Cash flows are discounted at a WACC of 8.4% over 10 years (Appendix H). It answers two questions, in black and orange:

Exhibit 8: The 1125R lost money, and keeping Buell loses money too, in every scenario

Net present value, $ million (10 years, 8.4% discount rate)



Bear: 5% margin, flat volume. Base (Harley's hope): 5% → 10% → 12.5%, +5% a year. Bull: up to 15%, +10% a year. Source: case pp. 6–7; author DCF.

How to read Exhibit 8: two questions, same bikes, different investment (base case, $m)

	Black: was the 1125R worth building?	Orange: is Buell worth keeping now?
Where we stand	Early 2007, before the $30m is spent	2008, after the $30m is spent
What is valued	One model, the 1125R, sold for 10 years and never replaced	Buell as a going business that keeps launching new models
Profit from selling bikes, after marketing, dealer and admin costs and tax (same in both)	+11.2	+11.2
Investment in models	−30.0 once, less 8.3 tax saving = −21.7	$6m a year for new models = −25.7 (the $30m is sunk)
Net present value	**−10.5**	**−14.6**
What it proves	Building Buell's 1125R was a mistake	Keeping Buell is still a mistake

The 1125R was a poor investment. Its value is negative even in the bull case (−$1.2m), and in Harley's own base case it destroys $10.5m. Breaking even would need about 10,900 sales in year 1, almost twice Buell's US volume. The bear and base cases stay negative at any discount rate from 6% to 12%.

Keeping Buell is negative too, and the estimate is generous. To compete, Buell must keep launching models; we assume one 1125R-sized model every five years ($6m a year), while Suzuki and Yamaha update their flagship sport bikes every two years and replace them every four to five[18]. So the real cost is higher than the −$14.6m shown.

The crisis makes it worse. By September 2007 Harley's stock was down 20%, and in the downturn it "saw sales and revenue plummet". With capital scarce, every dollar must earn the most it can: 35 cents of gross profit under the Harley name, 5 cents at Buell. The crisis also makes the bear case the most likely: a loss of $23.5m.

> **The sale test.** Freeze investment and invite offers. Sell if the proceeds exceed the value of keeping Buell, which is below zero in every scenario. Sell rather than close, to keep the value of its patented engineering and its riders who would never buy a Harley.

17 Buell's most powerful sport bike, a $30 million investment launched two weeks before Richer joined.
18 Model histories of the Suzuki GSX-R1000 and Yamaha YZF-R1 (Bennetts BikeSocial; Wikipedia); see Appendix H.

PART II · 6 CONCLUSION

Harley does not have a brand problem but a relevance problem, and it can solve it with assets it already owns

Harley's customers are ageing and young adults are not replacing them. They already rate Harley a leadership brand, but only 16% feel it is for them: the dealership intimidates them, the chrome image is not theirs, and they believe a Harley costs three times its real price. The answer is to make Harley-Davidson itself young again, not to rely on Buell. We recommend seven moves, all within the CMO's 27 months:

#	Move	What Harley does
1	**A bike they recognise**	Group the dark models into one Harley-Davidson line, led by the Iron 883 at $7,499 from January 2009
2	**A price they believe**	Lead every advert and showroom with the real entry price: "from $6,699"
3	**A dealership that welcomes them**	Young-adult Garage Parties, test rides, younger advisers and trained staff, piloted in a few dealers first
4	**A community they can join**	Rider's Edge training with the first purchase and H.O.G.[19] rides for new riders
5	**A brand they see themselves in**	Films that show young riders on dark Harleys, as *Easy Rider* and *Terminator 2* did for the Boomers, and a young apparel collection with a free test ride
6	**Money and data behind it**	Raise Outreach marketing from 17% towards 30% as pilots prove results, protect the Core's events, and report sales by customer segment
7	**Sell Buell**	The 1125R destroyed $10.5m and keeping Buell is worth −$14.6m; in a crisis every dollar must go where it earns 35%, not 5%

Moves 1–6 need under $1 million of new capital and are funded from the existing marketing budget, so they hold up in the crisis. Success will show in four numbers: young adult share above 22%, the "can relate" score from 16% to 24% or more, the median buyer age (47) no longer rising, and no loss of Core customers.

Main uncertainties. Buell's real price and costs are not in the case (the DCF tests a wider range); whether films and apparel turn into test rides must be proved in the pilots; and the depth of the crisis. The main risks, dealer resistance and Core backlash, are handled by central funding of pilots and by leaving the Core's bikes and events untouched (Appendix E).

Next steps: the first 90 days

1. Report 2007 sales by customer segment to set the baselines for every measure.
2. Choose 5–10 pilot dealers and launch young-adult Garage Parties, test rides and the "from $6,699" message.
3. Brief the agency on the dark line, film placements and the young apparel collection with a test-ride offer.
4. Freeze new Buell investment and appoint an adviser to test buyer interest.

> The Core keeps its chrome bikes and its rallies; the next generation gets a Harley of its own. This is how Harley-Davidson extends its mission, to fulfil dreams through the experience of motorcycling, to the generation that will ride after the Boomers.

19 Harley Owners Group: Harley's riders' club, over 600,000 members in 2007.

APPENDIX

Supporting analysis

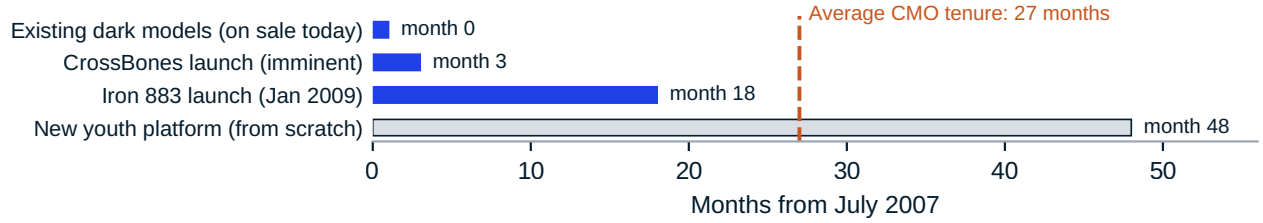
A. Prioritisation and timing of the options

Each option was scored on the four prioritisation criteria from class: impact, ability to influence, effort and strategic fit. The options in blue score well on all four. The new youth platform fails on ability to influence, because it would arrive too late; Buell as the gateway fails on impact and fit, because it does not make the Harley brand younger; emerging markets fail on impact for this problem, because they do not renew the US customer base.

Time decides the product question: a new platform would arrive 21 months after the average CMO has left.

Only existing and nearly finished products fit inside the CMO's window

Months from July 2007 until each product is available



CrossBones: a new dark, stripped-down Softail model, already developed and “soon-to-be-introduced”. Iron 883 debut: January 2009 (about month 18). Source: case pp. 1 and 6.

B. Barriers and levers at the dealership

Each barrier has a lever Harley already owns

Barrier	Evidence	Lever
Dealership aversion (largest)	Young adults find the visit daunting and intimidating	Extend the Garage Party format, proven with women, to young adults; offer evening events and test rides; recruit younger salespeople where useful and let customers choose an adviser they relate to; train all staff to welcome first-time riders
Price misperception	"Every Harley costs $20 grand" against an actual $6,699	Lead every message and showroom display with the entry price: "Sportster from $6,699", "Iron 883 at $7,499"
"Haven't seen the right bike"	Preference for dark, low-chrome bikes	Display the dark line as one collection at the front of the showroom
"May not belong"	16% of under-35s relate to the brand, against 32% of over-35s	Offer Rider's Edge training with the first purchase; create H.O.G. entry rides for new riders; build the message on belonging

Source: case pp. 4–6 and Harley-Davidson internal research, September 2007.

These changes should first be piloted in a small number of dealerships. With a recession starting and dealers sceptical of Outreach spending, a large untested campaign would be a risky response. A pilot creates the evidence dealers need: visits, test rides and conversion rates for under-35s, measured against the rest of the network.

Rebalancing marketing effort

Only 17% of planned US marketing in 2007 targeted Outreach groups, while the case states that attracting young adults is the brand's biggest open question. The share should rise in stages towards about 30%,[20] funded within the existing budget and released only as pilots prove conversion. The Core's rallies, H.O.G. support and dealer events are ring-fenced so that the shift does not read as abandonment.

20 Author assumption; the case gives no target. The final figure should be set from pilot results.

Size of the prize

The case does not give the size of the young adult market, so the prize is shown per unit of success. Every 1,000 young adults who buy an Iron 883 bring about $7.5 million of revenue and about $2.6 million of gross profit at the brand-average margin.[21] Buell would need about $52 million of revenue at its 5% margin to earn the same gross profit. The larger prize is lifetime value: a 25-year-old rider brought into the brand can trade up within it for decades, as the Boomers did.

Films and apparel. The case says the Boomers bought Harleys "in fulfillment of dreams fed by films like *Easy Rider* and *Terminator 2*". Films can do the same for the next generation, provided they show young riders on the dark bikes rather than older men on chrome choppers, the image young adults do not relate to. Clothing is already a sign of belonging in the Harley community; a young collection in the dark style is a low-cost first contact with the brand, much as Hard Rock Café T-shirts were worn by people with little interest in the restaurants. To make sure apparel leads to motorcycles, each young-collection purchase includes a free test ride or a Rider's Edge discount. Both are funded from the shift of marketing towards Outreach and judged on test rides and the "can relate" score, not on merchandise sales.

C–D. Implementation plan and success measures

Implementation plan

Timing	Actions	Decision gate
Months 0–3: diagnose	Report sales by customer segment, not only by model family; survey young visitors and non-buyers; measure perceived entry prices; select pilot dealers; freeze new Buell investment	Baselines set for conversion and price perception
Months 3–9: pilot	Launch the dark line in pilot dealers with entry-price messaging; young-adult Garage Parties, test rides and Rider's Edge bundles; test cultural and apparel-led promotion; open the Buell sale process	Pilot dealers beat the network on under-35 visits and conversion
Months 9–18: scale	Roll out proven levers nationally; raise the Outreach share of marketing in stages; launch CrossBones; prepare the Iron 883 launch	Cost per young-adult sale within target; Core retention stable
Month 18: launch	Iron 883 (January 2009) as the entry hero of the dark line at $7,499	Iron 883 passes its payback of about 380 bikes
Months 18–27: review	Review KPIs; decide on Buell bids against its value inside the group	Sell Buell only if the offer passes the test in Section 5

Source: author analysis; dates from the case.

Success measures

Measure	Baseline (2007)	Proposed direction or target
Harley share of Young Adult on-road buyers	22%	Rising year on year; target set from the pilot baseline
"You can relate to Harley-Davidson", under-35s	16%	Halve the gap to over-35s: 24% or more[a]
Median age of Harley buyers	47	Stop rising
Share of US marketing spent on Outreach	17%	About 30%, released in stages[a]
Under-35 dealer visit-to-purchase conversion	Not measured	Measured from month 3; pilot dealers above the network
Iron 883 units sold	Launch 2009	Above about 380 bikes (development payback)
Core retention and H.O.G. renewals	600,000+ members	No decline attributable to Outreach

Source: baselines from the case (pp. 4–6). [a] Proposed targets are author assumptions.

21 Assumes the case's 35% average gross margin applies to the Iron 883 at its retail price. The dealer price is lower, so these figures are an upper bound.

E. Risks and what-ifs

What if…	Response
The recession deepens	The plan relies on existing bikes and an investment of less than $1 million; the lowest-priced models also suit tighter budgets. Pilots can pause without stranded costs.
Dealers resist the changes	Fund the pilots centrally and show the conversion data before asking dealers to invest.
Core customers feel abandoned	Keep the chrome line, rallies and H.O.G. support unchanged, and present the dark line as a new family, not a new Harley.
Young adults still choose sport bikes	Some will. Harley does not need speed buyers; it needs the share who want authenticity and community, which its research shows exists.
The Iron 883 is delayed	The existing dark models carry the line; the Iron 883 strengthens it but the plan does not depend on it.
No buyer pays enough for Buell	Keep Buell without new models and re-run the DCF against the 10% margin target. If the bear case applies, plan an orderly wind-down.

What would change the recommendation

Keeping and investing in Buell would become right if credible forecasts showed attractive future cash flows, real incremental customers who would never buy a Harley, or offers well below Buell's value inside the group. Cultural marketing should be cut back if it generates apparel sales and publicity but no improvement in motorcycle consideration or dealer conversion.

F. Assumptions and information from outside the case

Item	Treatment
Mission statement and values	Not in the case; taken from secondary sources reporting Harley-Davidson's statements of the period
Iron 883 gross margin	Brand-average 35% applied to the $7,499 retail price; an upper bound because the dealer price is lower
Outreach share target (about 30%) and "can relate" target (24%+)	Author proposals; the case gives no targets
Hard Rock Café apparel analogy	Not in the case; used as an illustration of apparel as a first brand contact, not as evidence
Supplier power	Not discussed in the case; treated as not decisive
Buell DCF inputs (price, operating costs, tax, discount rate, product spending)	Author assumptions, listed in Appendix H and stress-tested; Buell's sale value is unknown

G. Key calculations

Figure	Calculation
2007 decline: Harley vs industry	Harley 562.9k − 311.6k others = 251.3k; (251.3 − 267.7) ÷ 267.7 = −6.1%; industry (562.9 − 586.7) ÷ 586.7 = −4.1%
Rate of ageing of the customer base	(47 − 35) ÷ (2007 − 1987) = 0.6 years per year
Price perception gap	$20,000 ÷ $6,699 = 3.0x
Iron 883 gross profit and payback	$7,499 × 35% = $2,625; $1,000,000 ÷ $2,625 ≈ 381 bikes (upper bound, as cost is "less than $1 million")

Size of the prize per 1,000 Iron 883 buyers	1,000 × $7,499 = $7.5m revenue; × 35% = $2.6m gross profit; Buell equivalent: $2.6m ÷ 5% = $52m revenue
Product timing	Iron 883: July 2007 to January 2009 = 18 months; new platform: 48 − 27 = 21 months after the average CMO leaves

I. Sources

Roese, N. J. & Kompella, M. (2013). *Harley-Davidson: Chasing a New Generation of Customers*. Kellogg School of Management, case KEL742, pp. 1–8 and Exhibits 1–5.
Harley-Davidson mission and values of the period, as reported in secondary sources: slideshare.net/slideshow/harley-davidson-51623017; qualitydigest.com/node/9850.
Course frameworks: problem definition and the six questions, mission elements, Porter's Five Forces and the STP model, decomposition trees and prioritisation criteria (Applied Problem-Solving, Sessions 1–3 and 6).
This report uses only information available in 2007–08 and does not draw on later outcomes.

H. Buell discounted cash flow

Input	Value	Basis
Year-1 volume	5,600 bikes	Case: "fewer than 5,600 Buells domestically" (US only). The 2007 10-K shows 3,700 US registrations, so this flatters Buell
Gross margin	5% → 10% → 12.5% (base)	Case: 5% today, 10% hoped for "within the next year", "strengthened beyond that as volumes grew"
Price per bike	$7,499	Assumption: no Buell price in the case; same as the Iron 883 for comparability; tested at $10,000–15,000
Operating costs	6% of revenue	Consistent with the case: Buell "rarely showed a profit" at a 5% gross margin; tested at 4–8%
Tax	35%	Harley's 2007 effective rate was 35.5% (10-K); losses offset other group profits
Horizon	10 years, no terminal value	Life of a sport-bike model; working capital ignored (small, and would lower NPV further)

H.1 Discount rate: a simple WACC of 8.4%

Component	Value	Basis
Risk-free rate (Rf)	5.0%	US 10-year Treasury yield, July 2007 (4.96%)
Equity risk premium (ERP)	4.5%	Between the implied US premium of early 2007 (about 4.2–4.4%, Damodaran) and long-run historical averages
Beta (β)	1.0	Assumption: market-average risk
Cost of equity = Rf + β × ERP	**9.5%**	CAPM
Cost of debt, before tax	6.0%	Assumption: about 1 point above the 2007 Treasury yield, for an investment-grade company
Tax rate	35%	Harley's 2007 effective rate was 35.5% (10-K)
Weights: equity / debt	80% / 20%	Simple assumption for the capital structure
WACC = 80% × 9.5% + 20% × 6.0% × (1 − 35%)	**8.4%**	Rate used to discount all cash flows; tested from 6% to 12% in H.4

Sources: macroradar.io (10-year Treasury, 24 July 2007); A. Damodaran, implied equity risk premiums (NYU Stern); Harley-Davidson 2007 Form 10-K (effective tax rate). Beta, cost of debt and weights are author assumptions.

H.2 Was the $30 million 1125R a good investment? Base case (Harley's own margin hopes)

Year 0 = 2007; $ million unless stated; negative values in brackets

Year	0	1	2	3	4	5	6	7	8	9	10
OPERATING DRIVERS											
Units sold		5,600	5,880	6,174	6,483	6,807	7,147	7,505	7,880	8,274	8,687
Price per bike ($)		7,499	7,499	7,499	7,499	7,499	7,499	7,499	7,499	7,499	7,499
Gross margin		5.0%	10.0%	10.0%	12.5%	12.5%	12.5%	12.5%	12.5%	12.5%	12.5%
INCOME STATEMENT ($M)											
Revenue		42.0	44.1	46.3	48.6	51.0	53.6	56.3	59.1	62.0	65.1
Gross profit		2.1	4.4	4.6	6.1	6.4	6.7	7.0	7.4	7.8	8.1
Operating costs (6% of revenue)		(2.5)	(2.6)	(2.8)	(2.9)	(3.1)	(3.2)	(3.4)	(3.5)	(3.7)	(3.9)
Depreciation		(6.0)	(6.0)	(6.0)	(6.0)	(6.0)	–	–	–	–	–
EBIT		**(6.4)**	**(4.2)**	**(4.1)**	**(2.8)**	**(2.7)**	**3.5**	**3.7**	**3.8**	**4.0**	**4.2**
Tax at 35% (credit when EBIT < 0)		2.2	1.5	1.5	1.0	0.9	(1.2)	(1.3)	(1.3)	(1.4)	(1.5)
NOPAT		**(4.2)**	**(2.8)**	**(2.7)**	**(1.8)**	**(1.7)**	**2.3**	**2.4**	**2.5**	**2.6**	**2.8**
FREE CASH FLOW ($M)											
+ Depreciation (non-cash)		6.0	6.0	6.0	6.0	6.0	–	–	–	–	–
− Investment	(30.0)	–	–	–	–	–	–	–	–	–	–
Free cash flow	**(30.0)**	**1.8**	**3.2**	**3.3**	**4.2**	**4.3**	**2.3**	**2.4**	**2.5**	**2.6**	**2.8**
Discount factor at 8.4%	1.000	0.923	0.851	0.786	0.725	0.669	0.617	0.569	0.525	0.485	0.447
Present value	(30.0)	1.7	2.8	2.6	3.0	2.8	1.4	1.4	1.3	1.3	1.2
Cumulative present value	**(30.0)**	**(28.3)**	**(25.6)**	**(23.0)**	**(19.9)**	**(17.1)**	**(15.7)**	**(14.3)**	**(13.0)**	**(11.8)**	**(10.5)**

Net present value	**−$10.5m** · IRR about 0% against a 8.4% cost of capital: the project barely returns its $30m before discounting · discounted payback: not reached in 10 years

Link to Exhibit 8: the present value of years 1–10 is +19.5 = profit from selling bikes (+11.2) + tax saving on depreciation (+8.3); minus the $30.0m investment = **−10.5**. Each year's free cash flow combines both, e.g. year 1: NOPAT (4.2) + depreciation 6.0 = 1.8. Capex depreciated straight-line over five years; no terminal value, because the project ends with the model.

H.3 Should Harley keep Buell? Base case from 2008, with the product spending needed to stay competitive

Year 0 = 2007; $ million unless stated; negative values in brackets

Year	0	1	2	3	4	5	6	7	8	9	10
OPERATING DRIVERS											
Units sold		5,600	5,880	6,174	6,483	6,807	7,147	7,505	7,880	8,274	8,687
Price per bike ($)		7,499	7,499	7,499	7,499	7,499	7,499	7,499	7,499	7,499	7,499
Gross margin		5.0%	10.0%	10.0%	12.5%	12.5%	12.5%	12.5%	12.5%	12.5%	12.5%
INCOME STATEMENT ($M)											
Revenue		42.0	44.1	46.3	48.6	51.0	53.6	56.3	59.1	62.0	65.1
Gross profit		2.1	4.4	4.6	6.1	6.4	6.7	7.0	7.4	7.8	8.1
Operating costs (6% of revenue)		(2.5)	(2.6)	(2.8)	(2.9)	(3.1)	(3.2)	(3.4)	(3.5)	(3.7)	(3.9)
Depreciation		(6.0)	(6.0)	(6.0)	(6.0)	(6.0)	(6.0)	(6.0)	(6.0)	(6.0)	(6.0)
EBIT		**(6.4)**	**(4.2)**	**(4.1)**	**(2.8)**	**(2.7)**	**(2.5)**	**(2.3)**	**(2.2)**	**(2.0)**	**(1.8)**
Tax at 35% (credit when EBIT < 0)		2.2	1.5	1.5	1.0	0.9	0.9	0.8	0.8	0.7	0.6
NOPAT		**(4.2)**	**(2.8)**	**(2.7)**	**(1.8)**	**(1.7)**	**(1.6)**	**(1.5)**	**(1.4)**	**(1.3)**	**(1.1)**
FREE CASH FLOW ($M)											
+ Depreciation (non-cash)		6.0	6.0	6.0	6.0	6.0	6.0	6.0	6.0	6.0	6.0
− Investment	–	(6.0)	(6.0)	(6.0)	(6.0)	(6.0)	(6.0)	(6.0)	(6.0)	(6.0)	(6.0)
Free cash flow	**–**	**(4.2)**	**(2.8)**	**(2.7)**	**(1.8)**	**(1.7)**	**(1.6)**	**(1.5)**	**(1.4)**	**(1.3)**	**(1.1)**
Discount factor at 8.4%	1.000	0.923	0.851	0.786	0.725	0.669	0.617	0.569	0.525	0.485	0.447
Present value	–	(3.9)	(2.3)	(2.1)	(1.3)	(1.2)	(1.0)	(0.9)	(0.7)	(0.6)	(0.5)
Cumulative present value	**–**	**(3.9)**	**(6.2)**	**(8.3)**	**(9.7)**	**(10.8)**	**(11.8)**	**(12.7)**	**(13.4)**	**(14.0)**	**(14.6)**
Net present value	**−$14.6m** · the $30m 1125R is sunk and excluded; every year needs a share of the next model										

Link to Exhibit 8: profit from selling bikes (+11.2) minus model spending after tax ($6m × 65% = $3.9m a year, present value −25.7) = **−14.6**. Product spending: one 1125R-sized model every five years, i.e. $6m a year; in steady state depreciation equals this spending. Buell is already spending to catch up with Suzuki and Ducati, so this spending is part of keeping it. It is a conservative assumption: Suzuki and Yamaha update their flagship sport bikes about every two years and replace them every four to five (GSX-R1000: new 2001, 2005, 2009, updated 2003, 2007, 2012; YZF-R1: new 1998, 2002, 2007, 2009, updated 2000, 2004, 2006, 2012; Bennetts BikeSocial, Wikipedia).

H.4 Sensitivity: in the bear and base cases the 1125R loses value at any discount rate from 6% to 12%

Net present value, $m

1125R investment	6%	8.4% (WACC)	9.5% (Ke)	11%	12%
Bear	−23.2	**−23.5**	−23.7	−23.8	−24.0
Base	−8.3	**−10.5**	−11.5	−12.6	−13.3
Bull	+2.5	**−1.2**	−2.8	−4.7	−5.8
Keep Buell (with product spending)	**6%**	**8.4% (WACC)**	**9.5% (Ke)**	**11%**	**12%**
Bear	−30.7	**−27.5**	−26.2	−24.6	−23.6
Base	−15.9	**−14.6**	−14.0	−13.3	−12.9
Bull	−5.1	**−5.3**	−5.3	−5.4	−5.4

Other changes (1125R NPV at 8.4%)	Bear	Base	Bull
Operating costs 4% / 8% of revenue	−19.9 / −27.1	−6.1 / −14.9	+4.1 / −6.6
Price $10,000 / $15,000 per bike	−24.1 / −25.3	−6.8 / +0.6	+5.6 / +19.2
Volume of 3,700 (2007 US registrations, 10-K)	−22.9	−14.3	−8.2
Year-1 volume needed to break even	Never	~10,900	~5,900

Scenarios: bear = 5% margin, flat volume; base = 5% → 10% (years 2–3) → 12.5% (from year 4), +5% volume a year; bull = 5% → 10% → 15% (from year 3), +10% a year. In the bear case a higher price makes losses larger, because each bike loses money.